

The Volume Expansion of Parallel Bodies, Brunn–Minkowski Concavity, and the Three-Dimensional Isoperimetric Inequality

Notes on Convex Geometry and Surface Differential Geometry

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Abstract

These notes organize the background needed for the proof of the three-dimensional isoperimetric inequality in Section 2.7 of *Bodies of Constant Width*. The focus is on strict convexity, C_+^2 convex bodies, Brunn–Minkowski concavity, the Gauss map (spherical image map), the Weingarten map and the shape operator, principal and normal curvatures, the area element of a parallel surface, the Gauss–Bonnet theorem, and the Euler characteristic. The central formula is

$$V(\phi_t) = V + St + Mt^2 + \frac{4\pi}{3}t^3,$$

where $\phi_t = \phi + tB(o, 1)$ is the exterior parallel body.

Lookup guide

Page numbers below refer to printed page numbers; PDF page numbers are added when useful.

- Martini–Montejano–Oliveros, *Bodies of Constant Width*: strict convexity is in Section 2.4, p. 20, PDF p. 31; Brunn–Minkowski concavity and the isoperimetric argument are in Section 2.7, pp. 27–29, PDF pp. 38–40.
- Schneider, *Convex Bodies: The Brunn–Minkowski Theory*: the spherical image map, the Weingarten map, principal curvatures, and the notation C_+^k are in Section 2.5, pp. 113–115, PDF pp. 137–139; the general Steiner formula is in Chapter 4, especially Section 4.2, pp. 211–223.
- Lee, *Introduction to Riemannian Manifolds*: the shape operator and principal curvatures of hypersurfaces are in Chapter 8, pp. 235–238, PDF pp. 246–249; the Gauss–Bonnet theorem is Theorem 9.7, p. 277, PDF p. 288.
- Lee, *Introduction to Topological Manifolds*: the Euler characteristic is discussed in Chapter 6, pp. 178–179, PDF pp. 195–196; its topological invariance appears near Corollary 10.25, p. 268, PDF p. 284.
- Diestel, *Graph Theory*: Euler’s formula for plane graphs is Theorem 4.2.9, p. 101, PDF p. 122.

1 Basic convex-geometric notation and strict convexity

1.1 Convex bodies and Minkowski addition

We work in Euclidean space $E^n \simeq \mathbb{R}^n$. Let $B(o, r)$ be the closed ball centered at o with radius r . For two sets $A, B \subset E^n$, their Minkowski sum is

$$A + B = \{a + b : a \in A, b \in B\}.$$

A convex body means a compact convex set. The exterior parallel body of ϕ is

$$\phi_t = \phi + tB(o, 1) = \{x + ty : x \in \phi, |y| \leq 1\}.$$

Geometrically, ϕ_t is obtained by attaching to each point of ϕ a closed ball of radius t and taking the union.

1.2 Strict convexity

Definition 1.1 (strict convexity) A closed convex set $\phi \subset E^n$ is strictly convex if $\partial\phi$ contains no nondegenerate line segment. Equivalently, every support hyperplane intersects ϕ in exactly one point.

Reference. *Bodies*, Section 2.4, p. 20: “A closed, convex set ϕ is strictly convex if ϕ does not contain line segments in its boundary or, equivalently, if every support hyperplane intersects ϕ at a single point.”

Strict convexity guarantees uniqueness of the boundary point with a prescribed supporting normal direction. Hence the reverse spherical image map is single-valued. If a body has a flat face, a whole face can share the same outward normal, and uniqueness fails.

However:

- The Brunn–Minkowski inequality itself does not require strict convexity; convex bodies are enough.
- The general Steiner polynomial is valid for arbitrary convex bodies, with no smoothness or strict convexity assumption.
- The calculation using N, dN, k_1, k_2 requires differentiability of the unit normal field, typically a C^2 boundary. Strict convexity is not the main analytic requirement for this differential calculation.

2 The meaning of C_+^2

In Schneider’s notation, a convex body K is of class C^k if its boundary hypersurface is a C^k regular submanifold. The stronger notation $K \in C_+^k$ means that $K \in C^k$ and the spherical image map, or Gauss map,

$$u_K : \partial K \rightarrow S^{n-1}$$

is a diffeomorphism. For $k \geq 2$, this is equivalent to nondegeneracy of the Weingarten map at every boundary point, or equivalently to all principal curvatures being nonzero.

For a convex body principal curvatures are nonnegative; hence in dimension three,

$$C_+^2 \iff \partial K \text{ is } C^2 \text{ and } k_1, k_2 > 0.$$

Thus C_+^2 is stronger than strict convexity. It says that the boundary is smooth and has positive curvature in all tangent directions.

Reference. Schneider, Section 2.5, pp. 113-115: p. 113 defines the spherical image map u_K and the Weingarten map $W_x = d(u_K)_x$; p. 115 defines C_+^k and gives the equivalent condition in terms of nonzero principal curvatures.

3 Brunn-Minkowski concavity

3.1 The Brunn-Minkowski inequality

Theorem 2.7.4 of *Bodies* states the Brunn-Minkowski inequality: if $\phi, \psi \subset E^n$ are convex bodies, then

$$V(\phi)^{1/n} + V(\psi)^{1/n} \leq V(\phi + \psi)^{1/n}.$$

Here V denotes n -dimensional volume.

The exponent $1/n$ reflects the homogeneity of volume:

$$V(\lambda K) = \lambda^n V(K), \quad V(\lambda K)^{1/n} = \lambda V(K)^{1/n}.$$

Minkowski addition is linear in sets, while volume has degree n . Taking the $1/n$ power produces the correct degree-one quantity. Thus the concave object is $V^{1/n}$, not V itself.

3.2 Why Theorem 2.7.1 is a concavity statement

Theorem 2.7.1 of *Bodies* says that for convex bodies ϕ, ψ ,

$$F(t) = V((1-t)\phi + t\psi)^{1/n}, \quad t \in [0, 1],$$

is concave.

Here is the proof. Define

$$A_a = (1-a)\phi + a\psi, \quad A_b = (1-b)\phi + b\psi.$$

For $0 \leq s \leq 1$,

$$(1-s)A_a + sA_b = (1-r)\phi + r\psi, \quad r = (1-s)a + sb.$$

By the Brunn-Minkowski inequality,

$$V((1-s)A_a + sA_b)^{1/n} \geq (1-s)V(A_a)^{1/n} + sV(A_b)^{1/n}.$$

Equivalently,

$$F((1-s)a + sb) \geq (1-s)F(a) + sF(b),$$

which is precisely concavity.

3.3 The exterior parallel body case

Set $\psi = \phi + B(o, 1)$. Since ϕ is convex,

$$(1-t)\phi + t(\phi + B(o, 1)) = \phi + tB(o, 1) = \phi_t.$$

Thus

$$t \mapsto V(\phi_t)^{1/n}$$

is concave. More generally, for any $t_1, t_2 \geq 0$ and $0 \leq s \leq 1$,

$$(1-s)\phi_{t_1} + s\phi_{t_2} = \phi_{(1-s)t_1 + st_2},$$

so Brunn-Minkowski implies concavity on $[0, \infty)$. If the function is twice differentiable, then

$$\frac{d^2}{dt^2} V(\phi_t)^{1/n} \leq 0.$$

Reference. *Bodies*, Section 2.7, pp. 27-29: p. 27 states Theorem 2.7.1 and applies it to exterior parallel bodies; p. 29 states Theorem 2.7.4 and proves Theorem 2.7.1 from it.

4 The Gauss map, the Weingarten map, and the shape operator

Let $\phi \subset E^3$ be a C^2 convex body and set $\Sigma = \partial\phi$. Let

$$N : \Sigma \rightarrow S^2$$

be the outward unit normal field. In convex geometry this map is called the spherical image map, and it is also known as the Gauss map.

At $p \in \Sigma$, its differential is

$$dN_p : T_p\Sigma \rightarrow T_{N(p)}S^2.$$

But

$$T_{N(p)}S^2 = \{v \in \mathbb{R}^3 : \langle v, N(p) \rangle = 0\} = T_p\Sigma.$$

Therefore dN_p can naturally be regarded as a linear map $T_p\Sigma \rightarrow T_p\Sigma$.

In Schneider's convex-geometric convention,

$$W_p = dN_p$$

is the Weingarten map, and the scalar second fundamental form is

$$II_p(v, w) = \langle W_p v, w \rangle.$$

This bilinear form is symmetric; hence W_p is self-adjoint and can be orthogonally diagonalized.

In Lee's Riemannian convention, the shape operator s is defined by the Weingarten equation

$$\tilde{\nabla}_X N = -sX.$$

In Euclidean space $\tilde{\nabla}_X N = dN(X)$, hence

$$W_p = dN_p = -s_p.$$

This is the sign difference between the convex-geometric convention and Lee's convention.

For example, for a sphere of radius R with outward normal, convex geometry gives principal curvatures $1/R$, while Lee's shape operator is $-(1/R)I$ and has eigenvalues $-1/R$.

Reference. Schneider, Section 2.5, p. 113: u_K is the spherical image map/Gauss map, and $d(u_K)_x =: W_x$ is the Weingarten map. LeeRM, Theorem 8.13, p. 236: the Weingarten equation for a hypersurface is $\tilde{\nabla}_X N = -sX$.

5 Principal curvatures, principal directions, and normal curvature

5.1 Principal directions and principal curvatures

Since $W_p : T_p\Sigma \rightarrow T_p\Sigma$ is self-adjoint, the spectral theorem gives an orthonormal basis e_1, e_2 of $T_p\Sigma$ such that

$$W_p(e_i) = k_i e_i, \quad i = 1, 2.$$

The directions e_i are called the principal directions, and the eigenvalues k_i are called the principal curvatures. In the convex-geometric convention used here, a convex surface has $k_i \geq 0$.

Reference. Schneider, Section 2.5, p. 114: the eigenvalues of the Weingarten map are the principal curvatures. LeeRM, Chapter 8, p. 238: the eigenvalues of the shape operator are principal curvatures and the corresponding eigenspaces are principal directions.

5.2 Normal curvature

Let $v \in T_p\Sigma$ be a unit tangent vector. The plane spanned by v and $N(p)$ intersects the surface in a plane curve near p . The signed curvature of this curve in the direction $N(p)$ is the normal curvature in the direction v . Equivalently,

$$\kappa_n(v) = II_p(v, v) = \langle W_p v, v \rangle, \quad |v| = 1.$$

If

$$v = \cos \theta e_1 + \sin \theta e_2,$$

then Euler's formula gives

$$\kappa_n(v) = k_1 \cos^2 \theta + k_2 \sin^2 \theta.$$

Thus k_1, k_2 are the extreme values of the normal curvature; they are the normal curvatures in the principal directions.

We define the mean curvature and Gaussian curvature by

$$H = \frac{k_1 + k_2}{2}, \quad K_G = k_1 k_2.$$

Reference. Schneider, Section 2.5, p. 115 gives Euler's theorem expressing curvature in an arbitrary direction as a weighted sum of principal curvatures. LeeRM, Chapter 1, pp. 4–5 gives the intuitive plane-section description of normal curvature, while Chapter 8, pp. 237–238 gives the formal definitions of principal, mean, and Gaussian curvatures.

6 The area element of a parallel surface

The exterior parallel surface is parametrized by

$$F_t : \Sigma \rightarrow \partial\phi_t, \quad F_t(p) = p + tN(p).$$

For $v \in T_p\Sigma$,

$$dF_t(v) = v + t dN_p(v).$$

Choose an orthonormal basis of principal directions e_1, e_2 . Then

$$dF_t(e_i) = e_i + tW_p(e_i) = (1 + tk_i)e_i.$$

Hence the length scale factors in the two principal directions are

$$1 + tk_1, \quad 1 + tk_2.$$

Therefore the area scale factor is their product:

$$dA_t = (1 + tk_1)(1 + tk_2)dA.$$

Expanding,

$$dA_t = (1 + t(k_1 + k_2) + t^2k_1k_2)dA = (1 + 2Ht + K_Gt^2)dA.$$

This is just the Jacobian determinant on the tangent plane. Without choosing principal directions, the same formula is

$$dA_t = \det(I + tW_p)dA.$$

Choosing principal directions diagonalizes W_p , making the determinant equal to $(1 + tk_1)(1 + tk_2)$.

7 Gauss-Bonnet and the Euler characteristic

7.1 Why $\partial\phi \cong S^2$

Let $\phi \subset E^3$ be a compact convex body with $o \in \text{int}\phi$. Every ray $o + \rho u$, $\rho \geq 0$, intersects $\partial\phi$ in exactly one point. Indeed, its intersection with ϕ is a bounded closed interval, and convexity prevents several separated intersection intervals. Therefore the radial projection

$$\partial\phi \rightarrow S^2, \quad p \mapsto \frac{p - o}{|p - o|}$$

is a homeomorphism. Hence

$$\partial\phi \cong S^2.$$

If $\phi \in C_+^2$, the Gauss map gives an even stronger diffeomorphism $\partial\phi \cong S^2$.

7.2 The Euler characteristic

For a finite CW decomposition, the Euler characteristic is

$$\chi(X) = \sum_{j \geq 0} (-1)^j c_j,$$

where c_j is the number of j -cells. For the sphere S^2 , one can use one 0-cell and one 2-cell, so

$$\chi(S^2) = 1 + 1 = 2.$$

Alternatively, triangulating S^2 by the boundary of a tetrahedron gives $V = 4$, $E = 6$, $F = 4$, and therefore

$$\chi(S^2) = V - E + F = 4 - 6 + 4 = 2.$$

Diestel's Euler formula for connected plane graphs,

$$n - m + \ell = 2,$$

is the graph-theoretic version of the same fact. Adding one point at infinity turns the plane into S^2 ; vertices, edges, and faces become 0-, 1-, and 2-cells. This is why the outer face must be counted.

7.3 The Gauss-Bonnet theorem

LeeRM Theorem 9.7 states that if M is a compact Riemannian 2-manifold, then

$$\int_M K dA = 2\pi\chi(M).$$

For $\Sigma = \partial\phi \cong S^2$, this yields

$$\int_{\Sigma} K_G dA = 2\pi\chi(S^2) = 4\pi.$$

8 The volume expansion of a three-dimensional parallel body

From the area element formula,

$$A(\partial\phi_t) = \int_{\partial\phi_t} dA_t = \int_{\Sigma} (1 + 2Ht + K_G t^2) dA.$$

Set

$$S = \int_{\Sigma} dA, \quad M = \int_{\Sigma} H dA.$$

Using Gauss-Bonnet,

$$A(\partial\phi_t) = S + 2Mt + 4\pi t^2.$$

The derivative of the volume of the exterior parallel body with respect to t equals the boundary area:

$$\frac{d}{dt}V(\phi_t) = A(\partial\phi_t).$$

Therefore

$$V(\phi_t) = V + \int_0^t (S + 2Ms + 4\pi s^2) ds = V + St + Mt^2 + \frac{4\pi}{3}t^3.$$

Putting

$$C = \frac{4\pi}{3},$$

we obtain the formula in *Bodies*:

$$V(\phi_t) = V + St + Mt^2 + Ct^3.$$

Here $C = 4\pi/3$ is the volume of the unit ball in E^3 , not the surface area of the unit sphere.

9 Deriving the three-dimensional isoperimetric inequality

Let

$$g(t) = V + St + Mt^2 + Ct^3, \quad F(t) = g(t)^{1/3}.$$

By Brunn-Minkowski, F is concave on $[0, \infty)$, hence

$$F''(t) \leq 0.$$

A direct calculation gives

$$F''(t) = \frac{1}{9}g(t)^{-5/3}\{3g(t)g''(t) - 2(g'(t))^2\}.$$

Thus

$$3gg'' - 2(g')^2 \leq 0.$$

Expanding,

$$3gg'' - 2(g')^2 = -2(S^2 - 3MV) + 2(9CV - MS)t - 2(M^2 - 3CS)t^2.$$

Setting $t = 0$ yields

$$S^2 - 3MV \geq 0.$$

Since this quadratic polynomial is nonpositive for all sufficiently large t , its leading coefficient must be nonpositive; hence

$$M^2 - 3CS \geq 0.$$

Combining the two inequalities,

$$S^4 \geq 9M^2V^2 \geq 27CSV^2.$$

Since $S > 0$,

$$S^3 \geq 27CV^2 = 36\pi V^2.$$

Therefore

$$S^3 \geq 36\pi V^2.$$

This is the three-dimensional isoperimetric inequality, with equality precisely for a ball.

Summary of the logical chain

The proof has two independent parts. First,

$$\text{Brunn-Minkowski} \implies V(\phi_t)^{1/3} \text{ is concave} \implies F''(t) \leq 0.$$

Second, smooth convex surface geometry gives

$$V(\phi_t) = V + St + Mt^2 + \frac{4\pi}{3}t^3.$$

The key local differential-geometric identity is

$$dA_t = (1 + tk_1)(1 + tk_2)dA.$$

It follows from the differential of the Gauss map: dN_p is a self-adjoint linear map on the tangent plane. Choosing its eigenvectors as principal directions diagonalizes the parallel map $F_t(p) = p + tN(p)$, so the two length scale factors are $1 + tk_1$ and $1 + tk_2$.